Cambodia Rice Yield Prediction: A Comparative Study of Machine Learning Models

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***Abstract*—Rice is the staple crop of Cambodia and a cornerstone of its agricultural economy. Accurately predicting rice yield is critical for food security planning and agricultural policy. This paper presents a comparative study of three machine learning models — Linear Regression, Logistic Regression, and a shallow Neural Network — applied to Cambodia rice yield data spanning 1990 to 2024. The dataset combines annual rice production statistics from FAOSTAT with climate variables (mean temperature and precipitation) from the World Bank. Two tasks are evaluated: a regression task predicting yield in kg/ha, and a classification task predicting whether yield is high or low relative to the historical median. Two complementary evaluation protocols are used: a chronological 80/20 split that tests *temporal extrapolation*, and a 5-fold cross-validation that estimates *interpolation* performance. Under cross-validation, Linear Regression achieves $\mathbf{R}^2 = 0.92$ and outperforms the Neural Network ($\mathbf{R}^2 = 0.87$); under the chronological split, both models exhibit a large generalisation gap that we trace to a distribution shift between training and test years rather than to model capacity. Classification metrics are reported with explicit reference to a trivial baseline because the chronological test set is class-degenerate.**

***Index Terms*—rice yield prediction, Cambodia, linear regression, logistic regression, neural network, PyTorch, agricultural data, cross-validation**

I. Introduction

Cambodia is one of Southeast Asia's major rice-producing nations, with rice cultivation covering over 2 million hectares annually and employing a large proportion of the rural population [1]. Fluctuations in rice yield directly affect food security, rural livelihoods, and national export revenue. Understanding and predicting yield trends from historical data and climate variables therefore has significant practical value.

Machine learning offers a data-driven approach to this problem. Unlike physics-based crop models that require detailed agronomic inputs, ML methods can learn patterns directly from historical records. Prior work has applied deep learning to large-scale crop yield prediction at the county level in the United States, where neural networks have outperformed linear baselines on rich, high-dimensional inputs [4]. We investigate the opposite end of the data spectrum: a small, country-aggregated time series (35 annual observations) where simple linear models are expected to be competitive or superior.

The objectives of this work are:

- To construct a merged dataset from FAOSTAT rice statistics and World Bank climate data.
- To train and evaluate three ML models on two predictive tasks (regression and classification).
- To compare the models under two evaluation protocols (chronological split and 5-fold CV) and analyse their disagreement.

II. Datasets

A. Data Sources

Three publicly available datasets were merged:

- **FAOSTAT Crops and Livestock Products** [1]: Annual Cambodia rice statistics (area harvested in ha, production in tonnes, yield in kg/ha) from 1990 to 2024.
- **World Bank Climate Portal — Temperature** [2]: Annual mean surface air temperature (°C) for Cambodia from 1901 to 2024.
- **World Bank Climate Portal — Precipitation** [3]: Annual mean precipitation (mm) for Cambodia from 1901 to 2024.

B. Merged Dataset and Feature Selection

After merging on year, the final dataset covers **35 years (1990–2024)**. Of the four candidate predictors initially considered, **production (tonnes) was excluded** because yield is defined as production divided by area harvested; including production would constitute direct target leakage. The retained features and target are listed in Table I.

TABLE I
Dataset Features and Targets (after dropping production).

Variable	Unit	Role
Area harvested	ha	Feature
Mean temperature	°C	Feature
Precipitation	mm	Feature
Yield	kg/ha	Target (regression)
High/Low yield	binary	Target (classification)

The binary classification label is derived by thresholding yield at the **training-set** median (computed on y_{train} only to avoid look-ahead bias): years above the median are labelled *High* (1), others *Low* (0). All features are standardised using training-set statistics before training (z-score normalisation).

C. Exploratory Analysis

Figure 1 (left) shows the strong upward trend in Cambodian rice yield over the period of study, and (right) the Pearson correlation matrix between features and yield. Area harvested correlates strongly with yield ($r = 0.98$) because both reflect the secular intensification of Cambodian agriculture. Climate variables show comparatively weak correlations with yield ($r \approx 0.30$).

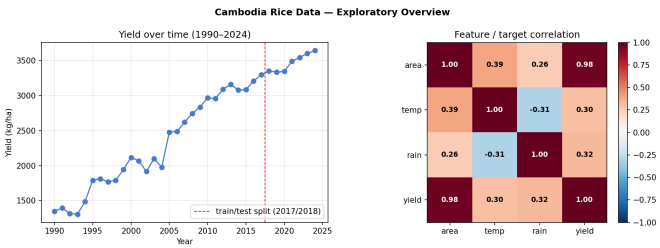

Fig. 1. Left: yield over time, with the chronological train/test split marked. Right: feature–target Pearson correlation matrix.

III. Proposed Methods

A. Linear Regression

Linear Regression models the yield as a linear combination of the three input features:

$$\hat{y} = \mathbf{w}^\top \mathbf{x} + b \tag{1}$$

The model is fitted using the ordinary least squares (OLS) solution provided by scikit-learn [5]. Linear Regression serves as a simple and interpretable baseline for the regression task.

B. Logistic Regression

For the classification task, Logistic Regression applies a sigmoid function to a linear combination of features:

$$\hat{p} = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \tag{2}$$

A threshold of 0.5 is used to assign the binary label. The model is implemented using scikit-learn's `LogisticRegression` class.

C. Neural Network

A shallow feedforward neural network is implemented in PyTorch [6] with the following architecture:

- **Input layer:** 3 neurons (one per feature)
- **Hidden layer:** 8 neurons with ReLU activation
- **Output layer:** 1 neuron
  - No activation for regression (MSE loss)
  - Sigmoid activation for classification (BCE loss)

Both variants are trained for 500 epochs using the Adam optimiser with a learning rate of 0.1. The architecture is intentionally kept small to limit overfitting on the 35-sample dataset.

IV. Experiments and Results

A. Experimental Setup

We use two complementary evaluation protocols:

- **Chronological split:** 80% training (28 samples, 1990–2017), 20% test (7 samples, 2018–2024). This protocol tests the model's ability to *extrapolate* into future years.
- **5-fold (stratified) cross-validation:** the dataset is shuffled (seed = 42) and partitioned into 5 folds; each fold serves as a held-out test set in turn. This protocol estimates *interpolation* performance and yields more stable estimates given $N = 35$.

For both protocols, feature standardisation and (for classification) the median threshold are fitted on each fold's training portion only, with no leakage from the held-out data. The neural network uses Adam (lr $= 0.1$), MSE/BCE losses, and 500 epochs; `torch.manual_seed(42)` ensures determinism.

B. Evaluation Metrics

Regression: MSE, RMSE (in kg/ha), and R^2. **Classification:** precision, recall, and F1-score. We additionally report two trivial baselines: (i) "predict the training mean" for regression and (ii) "predict the training majority class" for classification.

C. Results

TABLE II
Regression — chronological split (test = 2018–2024). Negative R^2 values indicate the model performs worse than predicting the test-set mean, which is a property of the small high-yield test window (see discussion).

Model	**RMSE** (kg/ha)	$\mathbf{R^2}$
Predict train mean (baseline)	1188.15	−96.86
Linear Regression	**250.50**	−3.35
Neural Network	452.87	−13.22

TABLE III
Regression — 5-fold CV (mean ± std).

Model	**RMSE** (kg/ha)	$\mathbf{R^2}$
Linear Regression	$\mathbf{163.65 \pm 30.41}$	$\mathbf{0.924 \pm 0.050}$
Neural Network	213.63 ± 91.89	0.875 ± 0.083

TABLE IV
Classification — chronological split. The test set contains 7/7 positive examples; metrics are degenerate (see discussion).

Model	**Precision**	**Recall**	**F1**
Logistic Regression	1.00	1.00	1.00
Neural Network	1.00	1.00	1.00

TABLE V
CLASSIFICATION — 5-FOLD STRATIFIED CV (MEAN ± STD).

Model	Precision	Recall	F1
Logistic Regression	1.000 ± 0.000	0.900 ± 0.200	$\mathbf{0.933 \pm 0.133}$
Neural Network	1.000 ± 0.000	0.850 ± 0.200	0.905 ± 0.131

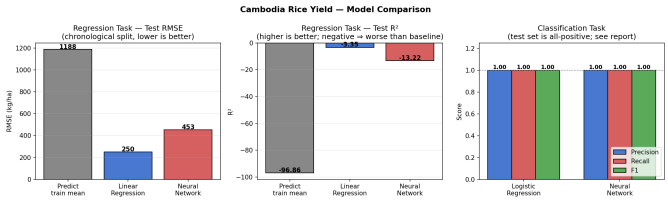

Fig. 2. Side-by-side model comparison on the chronological split: regression RMSE (left), regression R^2 (centre, negative bars descend below zero), and classification metrics (right).

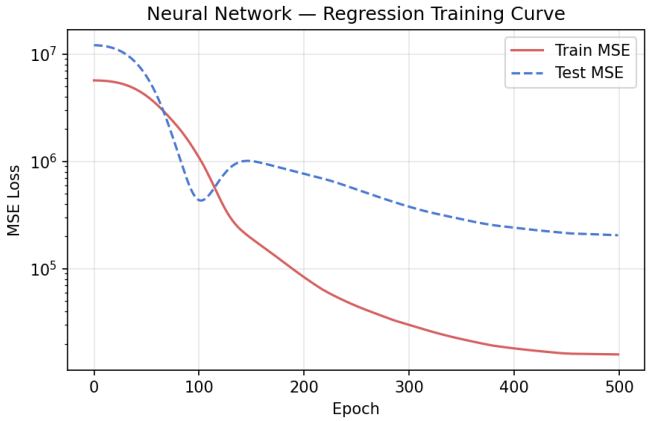

Fig. 3. Neural Network regression training curve. Train MSE drops to near zero while test MSE stalls above 2×10^5, indicating the network memorises the 28 training points without generalising to the 2018–2024 test window.

D. Discussion

Why is R^2 negative under the chronological split? The 7 test years (2018–2024) all lie above the maximum yield seen in training (test range 3335–3644 kg/ha versus train max 3297 kg/ha). The test set therefore lies entirely outside the training distribution, and any linear model trained on the earlier years will systematically under-predict. Negative R^2 here reflects this distribution shift, not a failure of the models per se: Linear Regression still beats the naive "predict train mean" baseline by a factor of nearly 5× in RMSE (250 vs. 1188 kg/ha).

Cross-validation tells a different story. Under 5-fold CV (where each fold's test points are interspersed with training points), Linear Regression reaches $R^2 = 0.924 \pm 0.050$. The neural network achieves $R^2 = 0.875 \pm 0.083$, which is competitive but slightly worse and substantially more variable across folds. This is consistent with theory: with $N = 35$, the bias-variance trade-off favours the simpler hypothesis class.

Standardised linear coefficients. The fitted Linear Regression weights (chronological split) are: area harvested $+621.0$, rainfall $+55.0$, temperature -18.1. Area harvested dominates the prediction — consistent with the very high yield–area correlation in Fig. 1 — while climate variables make small contributions. This is more a feature of country-aggregated time series (where mechanisation and irrigation effects load onto "area harvested") than evidence that climate is irrelevant at finer spatial scales.

Why are classification metrics perfect on the chronological split? The chronological test set contains only positive examples (all 7 years are above the train median). Any constant "predict High" predictor achieves F1 $= 1.0$ on this set, so the chronological classification numbers cannot distinguish between models. The 5-fold stratified CV provides a meaningful comparison: Logistic Regression (F1 $= 0.933 \pm 0.133$) edges out the Neural Network (F1 $= 0.905 \pm 0.131$), but the difference is well within one standard deviation.

Limitations. With $N = 35$ all conclusions remain noisy — the cross-validation standard deviations are large in absolute terms. A larger dataset (e.g., province-level annual yields) would tighten estimates and re-enable a fair extrapolation test.

V. CONCLUSION

We compared Linear Regression, Logistic Regression, and a shallow PyTorch neural network on Cambodia rice yield prediction (1990–2024). The headline findings are: (i) under interpolating cross-validation Linear Regression achieves $R^2 = 0.92$ and slightly outperforms the neural network, consistent with the bias-variance argument for simple models on small datasets; (ii) the chronological test split exhibits a strong distribution shift that drives both models to negative test R^2, but Linear Regression remains ∼5× better than the naive baseline; (iii) classification on the chronological split is degenerate because the test set contains a single class; the stratified-CV protocol restores meaningful comparison and again places Logistic Regression slightly ahead.

Future work should: (1) collect province-level yield data to expand the dataset by an order of magnitude; (2) incorporate features such as fertiliser use, flood occurrence, and irrigation coverage; and (3) explore recurrent architectures (e.g., LSTM) once enough data are available to support them.

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